Diffusion scores for Markov data: exact chain inference and estimation of the transition kernel

Giovanni Mantovani
Department of Computing Sciences
Bocconi University
`giovanni.mantovani@studbocconi.it`

Internal note, work in progress. Circulated for discussion, not submitted anywhere. Section 9 lists what is not yet supported; open decisions for the advisors are collected in `ANSWERS_AND_QUESTIONS_FOR_ADVISORS.pdf`. A claim-level audit against the implementation is in `REVISION_AUDIT.md`.

Abstract

Score-based generative models are trained by denoising regression, and the function they approximate — the posterior mean of the clean signal given its noised version — is unavailable in closed form for essentially every distribution of interest, so the error a network makes cannot be separated from the difficulty of the object it approximates. We study a setting where that separation is possible: a first-order Markov chain on $\mathbb{R}^n$ with continuous, non-Gaussian transitions. Coordinatewise Ornstein–Uhlenbeck noising multiplies the prior by unary factors, so the posterior remains a chain and sum-product on it is exact at the level of functional messages; a truncated grid turns this into a finite-state model for which the recursion is exact up to floating point, with truncation and quadrature errors measured separately. We then drop the assumption that the transition kernel is known and estimate it from noised sequences alone, within the linear-autoregressive family $K_\theta(a' \mid a) = \varphi_\theta(a' - \rho a)$ with both ρ and the innovation density free: the expectation step is exact for the discretised model and Fisher's identity supplies the gradient without differentiating the recursion. Under the correctly specified Markov model and at the budgets tested, the estimator attains lower pointwise denoising error than the neural baselines we trained. Through reverse diffusion the ranking differs at small innovation capacity; enlarging the mixture improves both axes monotonically, so in the range tested they do not trade off.

1 Introduction

Generative diffusion produces data $a \in \mathbb{R}^n$ from a distribution P_0 known only through samples, by noising and then reversing [1–3]. Reversal is driven by the score $S(x,t) = \nabla_x \log P_t(x)$, which is learned by regression. What that regression targets is the posterior mean of the clean signal, and for realistic data it is unknown, so a trained model cannot be scored against the thing it is approximating.

This note studies a family where the target is computable and not trivial. Distributions with tractable population scores are scarce, which is why the analytical literature leans on Gaussians and Gaussian mixtures; those are poor probes of structure, since the score is linear and any question about how a method exploits correlation has a degenerate answer. Hierarchical and discrete graphical models supply further tractable families [4–6], and the model here is a continuous, non-Gaussian member of the same class.

A first-order Markov chain gives a nonlinear score that remains exactly computable, for a structural reason: coordinatewise noising contributes one unary factor per site, which reweights node potentials without creating edges among the latent variables, so the posterior factor graph is the prior's chain.

We ask three questions. First, what does inference on the chain buy over a second-order Gaussian approximation, and where in the noise schedule does it matter. Second, whether the transition kernel can be estimated from noised data alone, and at what cost. Third, how the resulting estimator compares with trained denoisers, both pointwise and after reverse diffusion.

Contributions.

- A denoising reference for a continuous non-Gaussian correlated model (§3), with truncation and quadrature error separated and measured (§4), and the discretisation validated against a closed form and against brute-force enumeration.
- An identification of what single-Gaussian message closure computes on this model: exactly the LMMSE estimator of the covariance-matched Gaussian model (Proposition 1), proved directly, since a degree-two chain admits no large-degree central-limit argument.
- Estimation of the kernel — autoregressive coefficient and innovation density together — from noised sequences, with an expectation step that is exact for the discretised model and no differentiation through the recursion (§5).
- A pointwise comparison against neural baselines (§7) and a generative comparison (§8) whose rankings disagree at small innovation capacity, together with a capacity sweep showing that both axes improve monotonically as the innovation model is enlarged.

2 Model and the score–posterior identity

Notation. n is the chain length, which is also the ambient dimension, and M is the number of observed sequences. Much of the diffusion literature uses n for the sample count and d for the dimension; we index by sites because every graphical-model statement below is indexed by sites. K always denotes the Markov transition kernel and N_{g} always the number of quadrature points. A full symbol table is Appendix A.

Forward process. With η standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}x}{\mathrm{d}t} \;=\; -x + \sqrt{2}\,\eta(t), \qquad x(t) \;=\; e^{-t}a + \sqrt{\Delta_t}\,z, \qquad \Delta_t = 1 - e^{-2t}, \quad z \sim \mathcal{N}(0, I)\,. \quad (1)$$

The reverse-time convention, including the sign and the diffusion coefficient, is fixed once in Appendix A and used without variation.

Prior. P_0 factorises as

$$P_0(a) \;=\; \mu(a_1)\prod_{i=2}^{n} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \quad (2)$$

with φ an innovation density of mean zero and variance q, and $|\rho| < 1$. The implementation draws $a_1 \sim \mathcal{N}(0,1)$ and sets $q = 1 - \rho^2$.

Remark 1 (Covariance-stationary, not strictly stationary)**.** *With $q = 1-\rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly: $\mathrm{Var}(a_i) = 1$ for every i and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$, whatever the shape of φ. The chain is therefore* covariance-stationary. *It is **not** strictly stationary in distribution for non-Gaussian φ: the invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is the law of $\sum_{k\geq 0} \rho^k \varepsilon_{-k}$, which is not Gaussian, so drawing a_1 Gaussian starts the chain off the invariant law and the marginal of a_i drifts towards it over a burn-in of order $1/\log(1/\rho) \approx 6$ sites at $\rho = 0.85$. This does not affect the comparisons below, because every family shares the same a_1 law and the same covariance and so still differs only beyond second moments; but $q = 1 - \rho^2$ alone does not deliver strict stationarity, and we do not claim it does.*

Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly and differ only in higher moments.

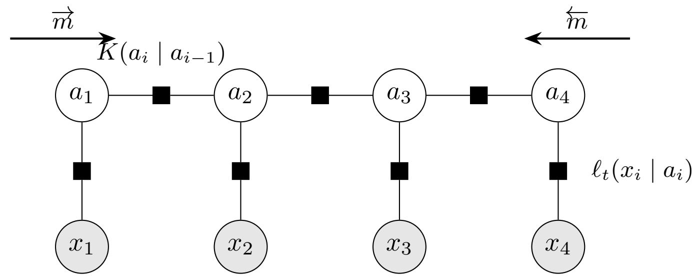

Figure 1: Factor graph of the posterior (4). Open circles are latent variables, shaded circles are observations, squares are factors. Each transition factor $K(a_i \mid a_{i-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor ℓ_t attached to a single latent variable. Conditioning therefore reweights node potentials without creating edges among the a_i, and the posterior graph remains a chain. Arrows show the two message sweeps (5).

The score is a posterior mean. From (1), P_t is a Gaussian convolution of P_0. Differentiating under the integral and dividing by $P_t(x)$ — the Gaussian factor over $P_t(x)$ is exactly the posterior density $P(a \mid x, t)$ — gives

$$S(x,t) \;=\; -\frac{x - e^{-t}\,\langle a\rangle_{x,t}}{\Delta_t}, \qquad \langle a\rangle_{x,t} = \int \mathrm{d}a\; a\, P(a \mid x, t). \tag{3}$$

Differentiation under the integral is justified whenever P_0 has a finite first moment (Appendix B). Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [7–9] and underlies denoising score matching [10, 11]. For squared loss, $\langle a\rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every denoiser below is scored against.

3 Inference on the noised chain

Conditioning (2) on x gives

$$P(a \mid x, t) \;\propto\; \mu(a_1) \prod_{i=2}^{n} K(a_i \mid a_{i-1}) \prod_{i=1}^{n} \ell_t(x_i \mid a_i), \qquad \ell_t(x_i \mid a_i) \propto \exp\!\left[-\frac{(x_i - e^{-t} a_i)^2}{2\Delta_t}\right]. \tag{4}$$

The likelihood factors are unary: the forward noise is independent across coordinates, so each observation touches one latent variable. They reweight node potentials and add no edges, and the posterior factor graph is the prior's chain (Figure 1). A chain is a tree, so sum-product returns the exact marginals [12–14]: with

$$\begin{aligned}
\overrightarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i-1}\; K(a_i \mid a_{i-1})\, \ell_t(x_{i-1} \mid a_{i-1})\, \overrightarrow{m}_{i-1}(a_{i-1}),\\
\overleftarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i+1}\; K(a_{i+1} \mid a_i)\, \ell_t(x_{i+1} \mid a_{i+1})\, \overleftarrow{m}_{i+1}(a_{i+1}),
\end{aligned} \tag{5}$$

initialised by $\overrightarrow{m}_1 = \mu$ and $\overleftarrow{m}_n \equiv 1$, the single-site marginal is $P(a_i \mid x, t) \propto \overrightarrow{m}_i(a_i)\ell_t(x_i \mid a_i)\overleftarrow{m}_i(a_i)$ and the pairwise marginal on edge $(i-1, i)$ is $\propto \overrightarrow{m}_{i-1}\ell_t(x_{i-1} \mid a_{i-1})K(a_i \mid a_{i-1})\ell_t(x_i \mid a_i)\overleftarrow{m}_i$. The local likelihood is absorbed *before* propagation, so $\ell_t(x_i \mid a_i)$ appears exactly once in each marginal; this convention is fixed throughout and pinned by a test. These are the forward–backward recursions of hidden Markov models [15, 16] written for a continuous state; when everything is Gaussian they reduce to the Kalman filter and RTS smoother [17, 18].

Remark 2 (Four statements, kept apart)**.** *(i) Sum-product on* (4) *is* exact at the level of functional messages*, as a consequence of the graph being a tree. (ii) Messages are functions on $\mathbb{R}$, so an implementation must represent them; a truncated uniform grid $[-A, A]$ with N_{g} points, spacing h and composite trapezoidal weights defines a* finite-state model*, for which the recursion is* exact up to floating-point arithmetic. *(iii) That finite-state model* approximates *the continuous one, with an error that must be measured. (iv) Any reported statistic of generated samples is in addition a* Monte

Carlo estimate. *We write "exact tree inference" for (i), "exact for the discretised model" for (ii), and "grid-BP reference" for the numerical arm; we never abbreviate any of these to "exact". In particular, the posterior marginals are not numerically exact merely because the graph is a tree.*

Naive marginalisation of (4) on a grid costs $O(N_{\mathrm{g}}^{n-1})$; the recursion costs $O(nN_{\mathrm{g}}^2)$ per sequence, each message update being one $N_{\mathrm{g}} \times N_{\mathrm{g}}$ matrix–vector product. This is the sense in which continuous BP is computationally operational: the messages are functions, and a finite representation of them turns the recursion into dense linear algebra whose cost is quadratic in the representation size and linear in the chain length.

4 Numerical control and the Gaussian second-order baseline

4.1 Separating truncation from quadrature

Two error sources are separated by construction. *Truncation*: mass outside $[-A, A]$ is discarded rather than transported. *Quadrature*: the trapezoidal rule is second order in h for integrands with bounded second derivative; $\varphi(e) \propto e^{-|e|/b}$ has a discontinuous derivative at the origin, so we claim no better than second order for the Laplace family and no spectral accuracy for it.

Figure 2 measures both. Panel (c) reports the edge-cell mass of the forward messages, taken as the worst site within a chain and then summarised over 256 chains per configuration. This is an *empirical diagnostic* of how much mass approaches the truncation boundary, not a bound on the truncation error, and the distinction is not pedantic: a single sampled trajectory — which is what an earlier version of this diagnostic reported — lands near the *median* over chains, while the maximum is some two orders of magnitude higher. We therefore plot both the maximum and the ninetieth percentile. The quantity that carries the argument is panel (d) together with the column-mass residuals, which involve no sample at all. Truncation responds to the domain and not the resolution: the residual falls from 3×10^{-4} at $A = 4$ to 1.4×10^{-8} at $A = 8$ and 1.4×10^{-10} at $A = 10$, essentially independently of N_{g}. Panel (d) reports the column-normalisation residual of the transition matrix restricted to interior columns, where no tail is truncated: at $A = 8$ it is 3.8×10^{-3}, 9.6×10^{-4}, 2.4×10^{-4} at $h = 0.08, 0.04, 0.02$, a factor of four per halving, which is the predicted second order. The two diagnostics are therefore doing different jobs, and quoting one convergence sweep alone would not have established the other. The working configuration $N_{\mathrm{g}} = 401$, $A = 8$ is chosen from these curves.

4.2 Validation where the answer is known

Take $\varphi = \mathcal{N}(0, q)$. Then a is jointly Gaussian with $(\Sigma_0)_{ij} = \rho^{|i-j|}$, and so is x, with $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$, giving

$$\langle a \rangle_{x,t} = e^{-t}\Sigma_0\Sigma_t^{-1}x, \qquad S(x,t) = -\Sigma_t^{-1}x. \tag{6}$$

The clean precision Σ_0^{-1} is tridiagonal, and the posterior precision $(e^{-2t}/\Delta_t)I + \Sigma_0^{-1}$ stays tridiagonal at every t: being Markov is a statement about the precision, not the covariance, which is dense. The Gaussian case fixes intuition — the score is a fixed linear map, so denoising is linear regression, which is exactly why a Gaussian prior is a poor probe of structure — and it supplies two independent checks.

At $N_{\mathrm{g}} = 401$, $A = 8$ the discretised recursion reproduces (6) to 9.2×10^{-15} relative, and agrees with brute-force enumeration of the discretised posterior — summing over all N_{g}^n configurations on a short chain, sharing no code with the message passing — to 1.6×10^{-14} on posterior means, 1.8×10^{-15} on the log-evidence and 3.9×10^{-16} on the pairwise statistic. Neither check involves non-Gaussian machinery: the pipeline is verified where the answer is known before being used where it is not. On the Laplace chain an independent importance-sampling estimate agrees with the recursion on the evidence to within five Monte Carlo standard errors at 10^6 samples and on the posterior mean to 10^{-2} relative at 2×10^6; these are test assertions whose tolerances are set by Monte Carlo error, not by the grid (`tests/test_em_bp.py`).

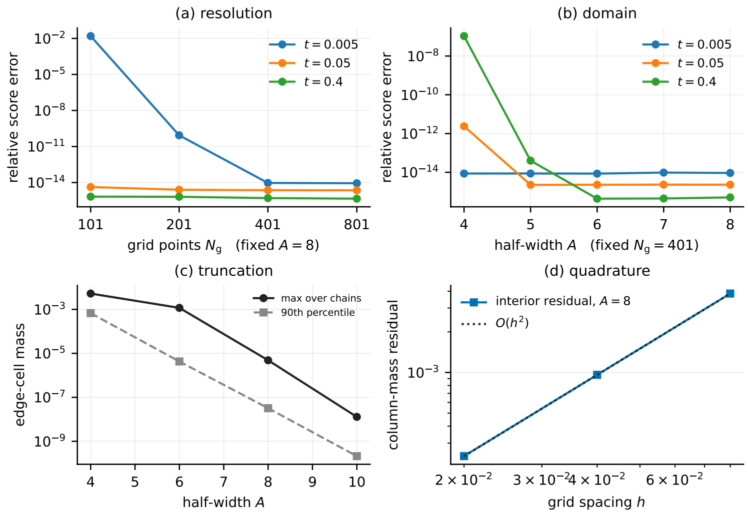

Figure 2: Grid and domain control, Laplace chain, $\rho = 0.85$, $n = 32$. (a) Relative score error against a high-resolution reference at fixed $A = 8$; (b) the same at fixed $N_g = 401$ as the domain grows. Both saturate at the floating-point floor $\sim 10^{-14}$, so beyond that the curves measure arithmetic, not discretisation. (c) Worst boundary mass over 32 sites, the truncation diagnostic, against A. (d) Interior column-mass residual against spacing at $A = 8$, the quadrature diagnostic, with an $O(h^2)$ reference. Panels (a)–(b) from `outputs/exp_01_grid_validation/grid_heatmap.csv`; (c)–(d) from `outputs/exp_18/boundary.csv`. All values are numerical, not empirical.

4.3 The Gaussian second-order baseline

A standard approximation propagates only the first two moments of each message. Under $a_i = \rho a_{i-1} + \varepsilon_i$, a message with mean m and variance v transforms as

$$\text{forward:} \quad m \mapsto \rho m, \quad v \mapsto \rho^2 v + q; \qquad \text{backward:} \quad m \mapsto m/\rho, \quad v \mapsto (v + q)/\rho^2, \tag{7}$$

the backward map being the forward one inverted because that recursion integrates over the output variable (it requires $\rho \neq 0$), and a Gaussian message times a Gaussian likelihood is Gaussian. The propagation uses only the mean and variance of φ.

Proposition 1. *For the chain* (2) *with linear transitions, Gaussian unary likelihoods and* $\mathbb{E}[a] = 0$, *moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{a}_{\text{LMMSE}} = \operatorname{Cov}(a, x)\,\operatorname{Cov}(x)^{-1} x = e^{-t}\Sigma_0 \left(e^{-2t}\Sigma_0 + \Delta_t I\right)^{-1} x, \tag{8}$$

for every innovation law with mean zero and variance q.

Proof. Both maps in (7) depend on φ only through its mean and variance, so the recursion returns the same function of x for every innovation law with moments $(0, q)$ — in particular the same as on the covariance-matched *Gaussian* chain. On that chain every message is exactly Gaussian, so the moment projection is the identity and the recursion returns the posterior mean. For jointly Gaussian zero-mean (a, x) the posterior mean is linear in x and therefore coincides with the best linear estimator under squared loss, which is (8). A full derivation is Appendix D. □

We call (8) the *Gaussian second-order baseline* and use that name throughout. On this model, message closure, moment projection and outright replacement of the prior by its covariance-matched Gaussian all coincide; they separate for richer message families, and the name is chosen not to prejudge which mechanism is at work. We do not call the method AMP: a degree-two chain supplies no large-degree central-limit justification for that terminology, and Proposition 1 is proved directly instead of argued asymptotically. The zero-mean hypothesis is load-bearing: with $\mathbb{E}[a] = 1.3$ the estimator (8) departs from the true LMMSE by 0.65 in maximum absolute value, because the general form carries $\mathbb{E}[a] + \operatorname{Cov}(a, x)\operatorname{Cov}(x)^{-1}(x - \mathbb{E}[x])$.

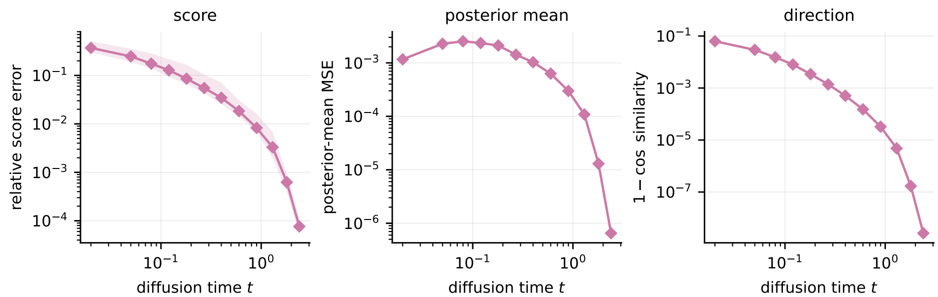

Figure 3: Error of the Gaussian second-order baseline against grid BP under the true kernel, Laplace chain, $\rho = 0.85$, $n = 40$, median over 60 trials with the 10–90 percentile band. Reference grid $N_{\mathrm{g}} = 801$, $A = 8$. Left: relative score error. Centre: posterior-mean MSE, which is non-monotone because it mixes the shrinking approximation error with the growing posterior variance. Right: one minus cosine similarity, which isolates direction from magnitude. Values are numerical approximations to the continuous model, aggregated over Monte Carlo trials. Source `outputs/exp_02_laplace_gaussian_message_error/laplace_summary.csv`.

The gap between (8) and grid BP locates where non-Gaussianity matters (Figure 3). On a Laplace chain at $\rho = 0.85$ the median relative score error falls from 0.175 at $t = 0.08$ to 7.6×10^{-5} at $t = 2.4$; large t Gaussianises the posterior, so the shape of φ survives only where the likelihood is weak. The same ordering holds across the other innovation families.

5 Estimating the transition kernel from noised sequences

We now treat K as unknown and observe only noised sequences x^μ, $\mu = 1, \ldots, M$, possibly at several noise levels. Write θ for the kernel parameters and $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(x^\mu)$.

What is and is not estimated. The kernel is estimated within the linear-autoregressive family

$$K_\theta(a' \mid a) \;=\; \sum_{c=1}^{C} \pi_c \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)\,, \tag{9}$$

in which **both** the autoregressive coefficient ρ and the innovation density are free. At $C = 4$ the free-parameter count is ρ (one), weights (four, one simplex constraint, so three), locations (four) and scales (four): **twelve**, against $3C$ in general. Every experiment initialises ρ at 0.3 against a truth of 0.85, so the autoregressive coefficient is genuinely recovered rather than supplied; from initialisations at $\rho^{(0)} \in \{0,\, 0.3,\, 0.6,\, -0.4\}$ the fitted value lands in $[0.8497, 0.8507]$ (Figure 4). What is *not* estimated is the initial law $\mu(a_1)$: it is held fixed at $\mathcal{N}(0,1)$, which is exactly the law the data-generating process uses, so there is no misspecification in μ anywhere in these results — but neither is there evidence that it could be recovered. The linear form of (9) is likewise assumed, not learned: what is free is a scalar and a density, not an arbitrary bivariate transition.

Expectation step. Many latent-variable models need an approximate E-step; this one does not, because the posterior is a tree and sum-product returns exact single-site *and pairwise* marginals — exactly for the discretised model, in the sense of Remark 2(ii). The complete-data log-likelihood is a sum over edges, so the θ-dependent part of the auxiliary function $\mathcal{Q}(\theta \mid \theta^{(r)})$ depends on the data only through

$$\Xi_{kl} \;=\; \sum_{\mu=1}^{M} \sum_{i=2}^{n} P\big(a_{i-1} = u_k,\; a_i = u_l \mid x^\mu\big)\,, \tag{10}$$

the expected transition mass between grid points, the continuum analogue of the expected transition counts of Baum–Welch [15]. Sufficiency requires a homogeneous kernel shared across sites and sequences, a fixed grid, an initial law excluded from θ, and θ-free likelihood factors — the last being what licenses observations at several noise levels to add into one Ξ. Given those, the maximisation step never revisits the data and costs $O(PN_\mathrm{g}^2)$ for P parameters, independently of M. Forming Ξ costs $O(MnN_\mathrm{g}^2)$ per iteration and $O(N_\mathrm{g}^2)$ storage, so "one matrix" is a statement about sufficiency, not about inference being cheap.

Gradient without differentiating the recursion. Fisher's identity [19–21] gives

$$\nabla_\theta \mathcal{L}(\theta) = \sum_\mu \mathbb{E}_\theta[\nabla_\theta \log P_\theta(a, x^\mu) \mid x^\mu] = \big\langle \Xi(\theta),\ \nabla_\theta \log K_\theta \big\rangle, \tag{11}$$

the exact gradient of the *marginal* log-likelihood from one sum-product pass. Sum-product is not a computation graph to backpropagate through; it supplies the conditional expectation. We verify (11) against finite differences of the discretised evidence to $\sim 10^{-9}$.

Which algorithm actually runs. Equation (11) licenses gradient ascent on $\mathcal{L}$; EM instead increases $\mathcal{Q}$. Both are implemented and neither dominates: on the smooth Gaussian kernel gradient ascent reaches EM's optimum to 2×10^{-9} nats while needing a tuned step size, and on the Laplace kernel it attains a higher likelihood, because there the exact maximiser of $\mathcal{Q}$ is a lattice artefact of the discretisation. The results below use EM. For the mixture kernel (9) the M-step is itself iterative — the complete-data problem carries a second latent variable, the mixture label, and we run four inner conditional-maximisation sweeps — so the algorithm is **generalised EM** of ECM type [22, 23]: each step increases $\mathcal{Q}$ without exactly maximising it, which preserves the monotone-ascent guarantee [24] but not the exact-maximiser property. We do not call it exact EM. Monotonicity is also observed: the largest decrease of $\mathcal{L}$ across iterations is 0 in all eight runs of Figure 4. Initialisation, stopping rule and parameter constraints are in Appendix F.

Innovation centering. The fitted mixture is *not* constrained to satisfy $\sum_c \pi_c \nu_c = 0$. Imposing it by reprojection after each update is not a maximisation step and broke monotone ascent by $\sim 10^{-2}$ nats in testing, so the condition is monitored as a diagnostic: it lands at -4.6×10^{-3} in the runs of Figure 4. The fitted family therefore does not satisfy the hypothesis of Proposition 1, which applies to the true kernel and only approximately to the estimated one.

Assumption 1 (Identifiability, assumed and not proved)**.** *Gaussian convolution at finite t is injective on distributions — the characteristic function acquires a factor $e^{-\Delta_t \|\xi\|^2/2}$, which has no zeros, and $e^{-t} > 0$ — so P_t determines P_0.* Kernel *identifiability does not follow. It additionally requires: the initial law fixed or separately identifiable; a full-support μ, since K is pinned down only on the support of μ; the family* (9) *identifiable as a parametric family, which for finite mixtures holds on the interior of the parameter set — all $\pi_c > 0$ and the pairs (ν_c, s_c) distinct [25, 26] — and is exactly where an over-specified C need not stay; and identification modulo label permutation. We assume these rather than prove them.*

What noising destroys. The channel does not degrade every coordinate of the kernel equally, and the disparity is large enough to govern what can be learned at all. Measuring it requires a parameter that is genuinely *higher order*: in the Gaussian model with parameters (ρ, q) both coordinates are second order, since q is a scale, so a comparison between them says nothing about shape. We therefore use a generalised-Gaussian innovation

$$\varphi_\beta(e) \ \propto\ \exp\big(-(|e|/s)^\beta\big), \qquad s(q,\beta) = \sqrt{q\,\Gamma(1/\beta)/\Gamma(3/\beta)},$$

in which β moves the fourth moment while holding the variance at q exactly: $\beta = 1$ is Laplace (excess kurtosis 3), $\beta = 2$ is Gaussian (excess kurtosis 0).

Comparing raw diagonal entries of the information matrix would ignore the coupling between coordinates, so we report the *efficient* information for each, obtained by projecting out the other two, $\mathcal{I}_{\beta\beta\cdot\eta} = J_{\beta\beta} - J_{\beta\eta} J_{\eta\eta}^{-1} J_{\eta\beta}$ with $\eta = (\rho, q)$, which is what bounds any estimator of β that must also fit the nuisance parameters. Between $t = 0.05$ and $t = 1.6$ the efficient information falls by 56–87 for the correlation, 235–522 for the variance, and 8.7×10^3–6.8×10^4 for the shape, so shape information decays 112 to 1222 times faster than correlation information depending on the true β (`outputs/exp_22/shape_information.csv`). The nuisance correction is itself asymmetric: at $t = 1.6$ the correlation and variance coordinates become 90% correlated and each loses a factor of about 5.5 to coupling, while β is nearly orthogonal to both and loses almost nothing. Higher-order structure of φ is thus far more fragile under the channel than second-order structure — by two to three orders of magnitude, not the factor of five a comparison within the Gaussian model would suggest — a fact §8 returns to.

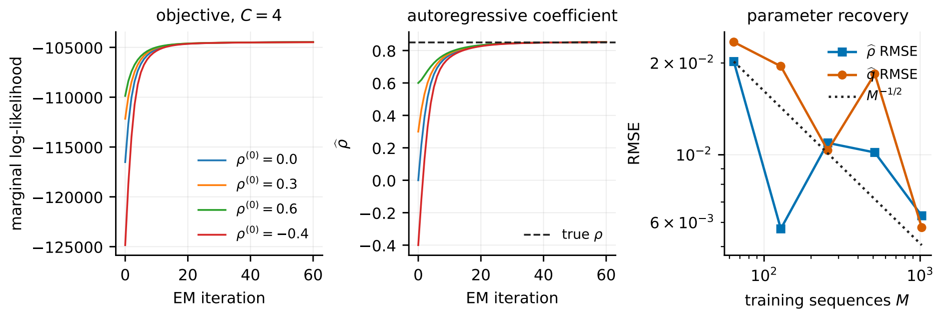

Figure 4: Optimisation diagnostics. Left: marginal log-likelihood by iteration at $C = 4$, $M = 512$, from four initialisations of ρ; the largest decrease across iterations is zero in all four, and in the four $C = 8$ runs not shown. Centre: the estimated autoregressive coefficient over the same runs, converging to $[0.8497, 0.8507]$ against a truth of 0.85 from initialisations spanning $[-0.4, 0.6]$ — direct evidence that ρ is estimated and not supplied. Right: RMSE of the recovered ρ and innovation variance against the number of training sequences, with an $M^{-1/2}$ reference. Sources `outputs/exp_18/em_trace.csv` and `outputs/exp_06_em_parameter_recovery/em_rate.csv`.

6 Experimental protocol

Unless stated otherwise: Laplace chain, $n = 32$, $\rho = 0.85$, grid $N_g = 401$, $A = 8$, five training noise levels $t \in \{0.1, 0.2, 0.4, 0.8, 1.6\}$, and 256 held-out sequences drawn from a seed disjoint from every training seed. Estimators are scored against grid BP under the *true* kernel on those held-out sequences.

Neural baselines are trained by denoising score matching in both standard parameterisations (predicting the noise, and predicting the clean signal). The fully connected network has 25,248 parameters; the locality-respecting baseline is a weight-shared one-dimensional convolution. Six replicates vary both the training set and the initialisation — the replicate index is mixed into every training seed — while the test set and its reference are common to all six, so the reported intervals cover sampling and optimisation variance together. Architectures, parameter counts, optimiser, step counts, batch sizes, seeds and hardware are in Appendix G; the reverse-integration protocol is in Appendix H.

Remark 3 (Information asymmetry)**.** *The estimator is given the Markov factorisation, the linear-autoregressive form of the kernel, and homogeneity across sites. The networks are given none of these. Every comparison below is therefore between a correctly specified structured model and generic function approximators, and should be read as measuring the value of that structure, not as a claim about neural denoisers in general.*

7 Pointwise denoising

Table 1 and Figure 5 show a ratio between 9 and 14 across seven doublings of the data, with no trend in the budget. The scoped statement this supports is: *under the correctly specified Markov model and at the data budgets tested, the structured estimator is more sample-efficient in pointwise denoising than the neural baselines we trained.* The network at $M = 4096$ did not reach the error the estimator attains at $M = 32$; we did not observe a crossing, so we do not quantify a data-equivalence factor, and the extrapolation required to do so spans several decades.

Controls. Capacity does not explain the networks' deficit: over 24 configurations from 3.2k to 297k parameters the best network reaches 0.208 against the estimator's 0.022 at $M = 1024$, and error degrades for the largest models. The baseline is competent: on a Gaussian chain, where the optimal denoiser is linear and closed-form, it passes the competence threshold asserted in `tests/test_em_bp.py`; the achieved value is not persisted, so we do not quote it.

Locality-respecting baseline. A fully connected network carries no sequential prior. Against a weight-shared window predictor the margin narrows from roughly 10 to between 2 and 4 over $M \in$

Table 1: Relative denoising error against grid BP under the true kernel, averaged over the noise schedule. Mean over six replicates, $\pm$ one standard error. The network column takes, at each noise level, the better of the two parameterisations. That is post-selection and favours the baseline, so we measured what it is worth rather than leaving it as a caveat: choosing the parameterisation on a disjoint validation bundle instead agrees with the test-set choice in 33 of 35 cells and changes the mean advantage from 11.65 to 11.66 — the oracle is worth $0.999\times$. The same holds for the harder two-axis case, where selecting receptive-field radius *and* parameterisation on validation — six configurations rather than two — agrees in 54 of 60 cells and is worth $0.996\times$. Reporting the noise-predicting parameterisation alone would overstate the network's deficit by 15–30%, since predicting the clean signal is better at $t \in \{0.8, 1.6\}$ at every budget. Sources `outputs/replicates/merged_summary.csv`, `outputs/exp_07_em_vs_score_network/sample_efficiency_val.csv`. Empirical, six replicates.

sequences M	network (25,248 params)	EM–BP (12 free)	ratio
32	0.6070 ± 0.0040	$\mathbf{0.0651 \pm 0.0031}$	9.3
128	0.4700 ± 0.0023	$\mathbf{0.0333 \pm 0.0035}$	14.1
512	0.2361 ± 0.0016	$\mathbf{0.0211 \pm 0.0022}$	11.2
1024	0.1736 ± 0.0009	$\mathbf{0.0182 \pm 0.0021}$	9.5
2048	0.1372 ± 0.0010	$\mathbf{0.0142 \pm 0.0009}$	9.7
4096	0.1240 ± 0.0003	$\mathbf{0.0124 \pm 0.0005}$	10.0

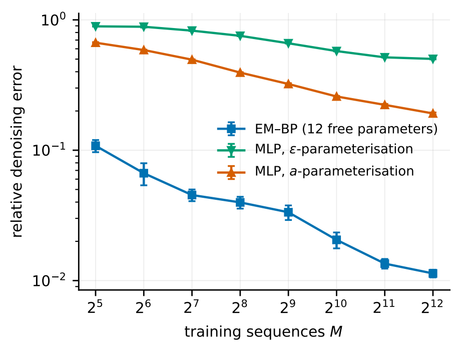

Figure 5: Relative denoising error against training-set size, mean over six replicates with standard errors. The two network curves are the two denoising-score-matching parameterisations reported separately rather than oracle-combined, so the gap to the estimator is visible without the post-selection of Table 1. The fitted innovation density itself is Figure 7 in Appendix I. Source `outputs/replicates/merged_summary.csv`. Empirical, six replicates.

$\{128, 512, 2048\}$, so 58–73% of the fully connected network's deficit was its architecture rather than the estimator. The margin against the convolution grows over those three budgets (2.31 ± 0.39, 3.36 ± 0.30, 4.16 ± 0.24 as ratios of seed-averaged errors; 2.72, 3.78, 5.04 if per-noise-level ratios are formed first), but three points do not establish a trend and the aggregation choice moves the value, so we report the range. **These numbers used a receptive field selected on the evaluation set** and are exploratory pending a rerun with a validation split; the bias favours the baseline, so the quoted margin is if anything conservative.

8 Reverse generation

Denoising accuracy is not what a generative model is judged on. Errors enter the reverse integration at every step and accumulate unevenly across the schedule, so a pointwise ranking need not survive. We integrate the reverse process of Appendix H under each estimator with common random numbers across estimators.

Behaviour near $t = 0$. Integration stops at $t_{\min} > 0$, and the reason is *conditioning* rather than a generic divergence of the score. Equation (3) carries $1/\Delta_t$, but its numerator vanishes at the same order: for a P_0 with a C^1 density $S(x,t) \to \nabla \log P_0(x)$, which is finite. Four effects are at work — a genuine singularity for the compactly supported uniform-innovation chain only, numerical cancellation, a likelihood factor the grid stops resolving, and a reverse drift whose Jacobian scales like $1/\Delta_t$ — and they are separated in Appendix H. The conclusion, $t_{\min} > 0$, is unchanged; the reason for it is not that S must blow up.

Comparing like with like. What each estimator produces is a sample of $P_{t_{\min}}$, not of P_0, so comparing it against clean data measures the stopping point rather than the score: independent Gaussian noise of variance v on an innovation of variance q multiplies the excess kurtosis by $(q/(q+v))^2$ (Appendix H). We instead draw a reference from P_0 and noise it forward to the same $t_{\min}$, so both sides are samples of the same law and the target follows in closed form. Run with the true kernel the sampler reproduces it to within 1.4% across budgets (medians 1.897, 1.883, 1.914, 1.908 at $M = 32, 128, 512, 2048$ against 1.9098), which validates the integrator and the statistics before any estimator is compared.

Sampler convergence. Over an eightfold range of step counts the reference arm's generated kurtosis is 1.989 ± 0.127, 1.817 ± 0.096, 1.913 ± 0.108, 1.936 ± 0.099 at 100, 200, 400, 800 steps: all within one standard error of the target, with the arm ordering unchanged throughout (Figure 8, Appendix I). The qualitative comparison is therefore not an artefact of one discretisation over that range. Two further step counts, 1600 and 3200, were launched but produced no output, so the sweep covers 100–800 and we claim nothing beyond it.

The rankings disagree at small capacity. At $C = 4$ the estimator is an order of magnitude more accurate pointwise (MSE against the Bayes denoiser 0.00052 against the convolution's 0.00525 and the fully connected network's 0.01330) and three times better on second-order structure (worst covariance-lag error 0.0292 against 0.0973 and 0.0518), and yet the convolution reproduces the *residual shape* more faithfully, by kurtosis (1.357 against 0.897, target 1.910) and by divergence (0.0028 against 0.0055).

Remark 4 (These residuals are not innovations)**.** *The statistic is the AR-filtered residual $r_i = x_i - \rho x_{i-1}$ of a sample of $P_{t_{\min}}$, not an innovation. Substituting $x_i = \alpha_t a_i + \sqrt{\Delta_t} z_i$ and $a_i = \rho a_{i-1} + \varphi_i$ gives $r_i = \alpha_t \varphi_i + \sqrt{\Delta_t}\,(z_i - \rho z_{i-1})$, so adjacent residuals are correlated,*

$$\mathrm{Cov}(r_i, r_{i+1}) = -\rho\Delta_t, \qquad \mathrm{Corr}(r_i, r_{i+1}) = \frac{-\rho\Delta_t}{\alpha_t^2 q + \Delta_t(1+\rho^2)},$$

which is -0.0997 at $\rho = 0.85$, $t_{\min} = 0.02$ and grows to -0.286 at $t_{\min} = 0.1$. The noised process is not AR(1); even in the Gaussian case it is ARMA-type. The marginal shape comparison above remains valid and remains the sharpest discriminator between families at matched covariance, but it does not measure recovery of the clean transition innovation law, and we do not claim that it does. On the clean chain the same filter does recover innovations (measured $|\mathrm{ACF}_1| < 0.02$).

The fully connected network shows the pattern most sharply, and under the corrected training protocol of §8 more sharply still: its generated residuals become statistically indistinguishable from Gaussian as data grows — excess kurtosis 0.046 at $M = 2048$, per-seed 0.069, 0.019, -0.002, 0.097 — while its pointwise error improves monotonically over the same range. One estimator moving in opposite directions on the two axes is the cleanest form this dissociation takes. It fits the bulk of the posterior and loses the tail.

It is not an artefact of the setting. The disagreement holds across all four non-Gaussian families at matched covariance (Table 3, Appendix I), is unchanged at $n \in \{32, 64, 128\}$, and is identical to three decimals under grid refinement $N_{\mathrm{g}} \in \{401, 801, 1601\}$, which rules out quadrature. On the *Gaussian* chain it disappears: the disagreement column of Table 3 is -0.028, within noise of zero and if anything favouring the estimator. That is the negative control the capacity account predicts and an information-theoretic one does not, since the channel degrades correlation information on a Gaussian chain as well: a four-component mixture represents a Gaussian innovation exactly, so there is no deficit to expose. Across the remaining families the size of the disagreement orders with how badly that mixture misrepresents the innovation, from 0.094 for the mildly bimodal case to 0.489 for the heaviest tail.

Both axes improve with capacity. Two explanations were available. Both the marginal likelihood and the squared-error loss are dominated by the bulk of the distribution, so the tail — little mass, high leverage on shape statistics — is weakly constrained by either; separately, the channel destroys innovation-shape information two to three orders of magnitude faster than correlation information (§5), so the data itself might not carry it. The first is a limit of the model class and is fixable; the second is a limit of the observation model and is not.

Varying C separates them (Table 2, Figure 6). Generated kurtosis climbs monotonically and pointwise error *also* improves monotonically, by a factor of 9.7 from $C = 2$ to $C = 16$: the two axes do not trade against each other anywhere in the range tested, and at $C = 16$ the estimator is ahead of the convolution on both at once.

Remark 5 (A third explanation, not controlled for: EM was stopped short)**.** *The two explanations above are not exhaustive. Every run in this section uses* `em_iters` $= 40$*, and EM on this kernel converges on ρ much faster than on the innovation* shape*: at $\rho = 0.85$, $M = 512$, $C = 8$ the fitted excess kurtosis is* 0.84 *at 30 iterations,* 2.30 *at 120 and* 2.29 *at 400, while ρ is flat from iteration* $\sim$25*. A converged-looking ρ trace is therefore not evidence of a converged kernel, and the estimator arm here may simply be underfitted.*

Rerunning this configuration (Laplace, composite, $M = 512$, grid 401*) with the iteration count as the only variable gives generated excess kurtosis* $0.625 \to 1.464$ *at $C = 4$ and* $0.871 \to 1.380$ *at $C = 8$ going from 40 to 200 iterations, against a convolution at* 1.367*. Two consequences: the capacity gain in fitted kurtosis falls from* 0.542 *to* 0.022*, so it is mostly a difference in convergence* rate *at a fixed budget rather than in capacity; and the ranking that opens this subsection reverses. That evidence is a single seed at* 800 *generated chains and* 200 *integration steps, enough to withdraw the attribution but not to replace it. The sweep needs rerunning at convergence before this paragraph is relied on. Nothing in §5 or the sample-efficiency results is affected.*

Remark 6 (Where capacity stops paying, on a paired design)**.** *Separate medians over a handful of seeds cannot resolve the small differences at the top of this range, because most of the variance is the training draw and it is shared between capacities. We therefore fit every C on the* same *training set within a cell and score it on the* same *held-out bundle, so the contrast is within-dataset and its standard error is an order of magnitude below the effect. Held-out denoising MSE against the Bayes denoiser, six paired seeds, differences against $C = 1$: at $M = 512$ the gains are $+3.8 \times 10^{-5}$ ($C = 2$), -1.41×10^{-3} (4), -1.74×10^{-3} (8), -1.76×10^{-3} (16), with standard errors 1.3×10^{-5} to 1.1×10^{-4} (*`outputs/exp_16/paired_local/paired_contrasts.csv`*).*

Three consequences. $C = 8$ is the optimum: $C = 16$ differs from it by less than one standard error and at $M = 128$ is marginally worse. A two-component mixture is not *an improvement on a single Gaussian innovation, and at $M = 512$ is significantly worse than one. And none of this is a discretisation artefact — the narrowest fitted component is at least* 8.8 *grid spacings wide in all sixty cells, with the effective component count tracking the nominal one, so the mixture is genuinely using its capacity and every component is resolved.*

This is consistent with a capacity bottleneck at small C and inconsistent with the information account being the binding constraint there. It does not show that information is never binding — the curve has not saturated, and we have not measured how much further it can go.

9 Limitations and open questions

The model assumes exact first-order Markov structure. Under a non-Markov prior the estimator is misspecified in a way more data does not repair, and how much that costs turns out to depend sharply on *which* assumption is violated — so we measured it rather than leaving it as a caveat.

Two mechanisms, both with an exact reference. A shared global latent, $a = (y + \beta g)/\sqrt{1 + \beta^2}$ with $g \sim \mathcal{N}(0, 1)$, contaminates the prior with a rank-one term; weak long-range precision coupling, $Q = Q_{\mathrm{AR}(1)} + \gamma Q_{\mathrm{long}}$, does not. For Gaussian innovations both references are linear algebra; for the Laplace chain plus a global latent the reference is exact by conditioning on g and marginalising it with Gauss–Hermite, since the prior is Markov given g (Appendix I).

The estimator is remarkably robust to the rank-one violation and fails quickly under the other. Against the convolution, the ratio of relative score errors stays above 2.08 across the whole global-latent sweep up to $\beta = 1$, where half the marginal variance is a shared constant; with Laplace

Table 2: Both evaluation axes against the mixture component count, $M = 2048$, Laplace chain. Left block: median MSE against the Bayes denoiser over five noise levels and three seeds (`outputs/exp_16/cpoint_C*/pointwise.csv`). Right block: median generated innovation excess kurtosis over three seeds, target 1.910 (`outputs/exp_16/components_C*/generation.csv`). The network arms do not depend on C and are identical down the table, which is an internal consistency check rather than a result. Empirical.

	MSE against Bayes denoiser			generated excess kurtosis		
C	EM–BP	CNN	MLP	EM–BP	CNN	MLP
2	0.002281	0.005161	0.012854	−0.034	1.273	0.170
4	0.000510	0.005161	0.012854	0.812	1.273	0.170
8	0.000276	0.005161	0.012854	**1.319**	1.273	0.170
12	0.000249	0.005161	0.012854	**1.363**	1.273	0.170
16	**0.000234**	0.005161	0.012854	**1.487**	1.273	0.170

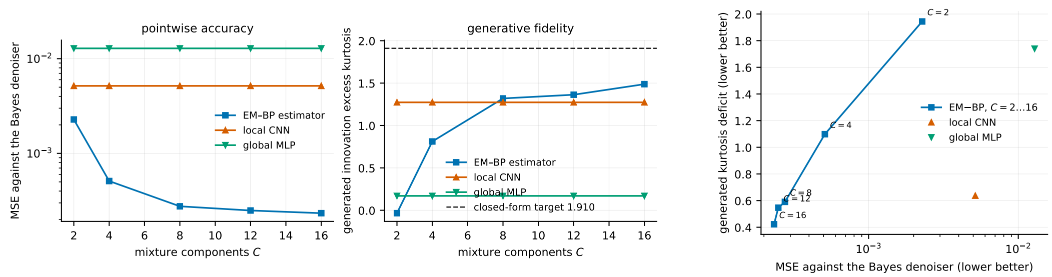

Figure 6: The capacity sweep on both axes. Left and centre: MSE against the Bayes denoiser (log scale) and generated innovation excess kurtosis against the closed-form target, both against C; the network arms are C-independent by construction. Right: the same data with one axis against the other, one point per configuration — both coordinates improve together, so the trace moves towards the origin rather than along a trade-off. Data as in Table 2. Empirical, three seeds; the sweep does not yet include marginal likelihood or runtime against C.

innovations it stays above 1.45. Under long-range coupling it inverts at $\gamma \approx 0.10$ and reaches 0.47 at $\gamma = 0.4$, where the fully connected network also overtakes the estimator. This is what the operator analysis of §4 predicts: a global latent makes the score residual exactly rank one, which a chain absorbs, while long-range coupling is not low-rank and cannot be represented. The fitted parameters corroborate it — EM converges to the best-Markov (Chow–Liu) projection to within 0.0009–0.0128 across the β sweep but departs from it by up to 0.101 under γ, exactly where the chain family stops being able to represent the truth. So the honest scope is: the structural prior survives rank-one contamination essentially intact, regardless of innovation shape, and should not be relied on under genuine long-range structure.

The transition family (9) is additionally misspecified at small C, which §8 measures and shows is relieved by enlarging C. The initial law is fixed rather than estimated, and the chain is covariance-stationary rather than strictly stationary (Remark 1); no run yet initialises from the invariant non-Gaussian law. Identifiability is assumed (Assumption 1), not proved. Sum-product is exact on the posterior chain but the implementation is a discretisation whose error, for the Laplace family, is second order rather than spectral.

Inference costs $O(nN_{\mathrm{g}}^2)$ per sequence, measured at 210–289 times a network forward pass; since reverse diffusion calls the denoiser at every step, this is the practical weak point. Training cost grows linearly in M while the network's is fixed by its step count, so the advantage is in data, not wall-clock. The low parameter count is bought with a structural assumption, and the advantage narrows as the parametric dimension grows: on a discrete-alphabet variant it falls from about 15 at six estimated parameters to 3.6–6.1 at 132.

The measurement gaps that remain are narrower than they were. The capacity sweep now includes marginal likelihood and runtime against C: held-out evidence saturates by $C \approx 8$ and cost is flat in C, since the $O(MnN_{\mathrm{g}}^2)$ expectation step dominates and only the maximisation step scales with capacity. Model selection is on a disjoint validation bundle throughout, and the oracle it replaces was measured at $0.999\times$ and $0.996\times$ (Table 1). The locality statement is now exact rather than a proxy: under strictly stationary initialisation the windowed estimator *is* $\mathbb{E}[a_i \mid x_{i-r:i+r}]$, because a contiguous window of a Markov chain is a chain with the correct endpoint law. That correction does not change the measured rates — the ratio of the stationary to the original protocol has median 1.004 across ninety cells — so the earlier numbers stand, on better footing.

What is still missing. The likelihoods reported here are a *composite* objective: each clean chain is observed at several noise levels and the groups are treated as independent, so every marginal is correctly specified but the effective sample size is the number of clean chains, not that times the number of levels. Identifiability remains assumed. The neural comparison is against the two architectures described — a fully connected network and a weight-shared convolution — and does not license a claim about modern sequence denoisers, which is the comparison a practitioner would want (Remark 3).

10 Conclusion

When a sequential distribution is known to have low-dimensional Markov structure, learning a diffusion score reduces to latent-variable inference plus estimation of a compact transition model. Sum-product on the noised chain supplies a non-Gaussian reference whose discretisation error is separately measured, and estimating the kernel from noised data alone gives a denoiser substantially more accurate at small sample sizes than the networks we trained, at the cost of a stronger structural assumption and much more expensive inference. Reverse diffusion does not reproduce that ranking at small innovation capacity, and enlarging the innovation model improves both axes together. The methodological point is the more portable one: an estimator can be an order of magnitude better at the task it was fitted for and still generate worse samples, so a structured method has to be judged on the generated distribution and not on denoising error alone.

References

[1] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International Conference on Machine Learning*, 2015.

[2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020.

[3] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[4] Antonio Sclocchi, Alessandro Favero, and Matthieu Wyart. A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 122(1):e2408799121, 2025.

[5] Jérôme Garnier-Brun, Marc Mézard, Emanuele Moscato, and Luca Saglietti. How transformers learn structured data: insights from hierarchical filtering, 2024.

[6] Song Mei and Yuchen Wu. U-Nets as belief propagation, 2024.

[7] Herbert Robbins. An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume 1*, pages 157–163, 1956.

[8] Koichi Miyasawa. An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4):181–188, 1961.

[9] Bradley Efron. Tweedie's formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614, 2011.

[10] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.

[11] Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709, 2005.

[12] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.

[13] Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger. Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519, 2001.

[14] Marc Mézard and Andrea Montanari. *Information, Physics, and Computation*. Oxford University Press, 2009.

[15] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970.

[16] Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

[17] Rudolph E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[18] Herbert E. Rauch, F. Tung, and Charlotte T. Striebel. Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450, 1965.

[19] Ronald A. Fisher. Theory of statistical estimation. *Mathematical Proceedings of the Cambridge Philosophical Society*, 22(5):700–725, 1925.

[20] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 44(2):226–233, 1982.

[21] Olivier Cappé, Eric Moulines, and Tobias Rydén. *Inference in Hidden Markov Models*. Springer, 2005.

[22] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38, 1977.

[23] Xiao-Li Meng and Donald B. Rubin. Maximum likelihood estimation via the ECM algorithm: a general framework. *Biometrika*, 80(2):267–278, 1993.

[24] C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11 (1):95–103, 1983.

[25] Henry Teicher. Identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 34(4): 1265–1269, 1963.

[26] Sidney J. Yakowitz and John D. Spragins. On the identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 39(1):209–214, 1968.

[27] Brian D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982.

[28] Ulrich G. Haussmann and Etienne Pardoux. Time reversal of diffusions. *The Annals of Probability*, 14(4):1188–1205, 1986.

[29] Erich L. Lehmann and George Casella. *Theory of Point Estimation*. Springer, 2nd edition, 1998.

[30] Steven M. Kay. *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall, 1993.

A Notation and diffusion conventions

A.1 Symbols

symbol	meaning	symbol	meaning
n	chain length = ambient dimension	$K,\ K_\theta$	transition kernel
M	number of observed sequences	φ	innovation density
N_{g}	number of quadrature grid points	ρ	autoregressive coefficient
$u_1, \dots, u_{N_{\mathrm{g}}}$	grid points	q	innovation variance
h	grid spacing	C	mixture components
A	grid half-width, domain $[-A, A]$	μ	initial law $\mu(a_1)$
t	diffusion time	ℓ_t	unary likelihood factor
$\Delta_t = 1 - e^{-2t}$	noise variance	$\overrightarrow{m},\ \overleftarrow{m}$	forward, backward message
$P_0,\ P_t$	clean, noised distribution	θ	kernel parameters
$S(x, t)$	score $\nabla_x \log P_t(x)$	$\mathcal{Q}$	EM auxiliary function
$\langle a \rangle_{x,t}$	posterior mean of a given x	Ξ	expected transition statistic
η	standardised Gaussian white noise	$\mathcal{L}$	marginal log-likelihood
$t_{\min},\ t_{\max}$	reverse-integration endpoints	$\widehat{a}_{\mathrm{LMMSE}}$	LMMSE estimator

Two symbols carry a history worth stating, because an earlier draft of this work used them differently. K **always means the transition kernel** and never a grid size; the grid size is always N_{g}. The ambient dimension is written n because it *is* the chain length and every graphical-model statement is indexed by sites; the diffusion literature more often writes n for the sample count, which here is M.

A.2 Time orientation and the reverse SDE

Convention, used without variation. η is standardised Gaussian white noise, $\langle \eta(t)\eta(t') \rangle = \delta(t - t')$, so every $\sqrt{2}$ is written explicitly.
Forward, t running from 0 to $t_{\max}$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -x + \sqrt{2}\,\eta(t), \qquad x(t) = e^{-t}a + \sqrt{\Delta_t}\,z, \qquad \Delta_t = 1 - e^{-2t}, \quad z \sim \mathcal{N}(0, I)\,.$$

Reverse, in the elapsed-time variable $\tau = t_{\max} - t$ running from 0 to $t_{\max} - t_{\min}$, so that increasing τ means decreasing t:

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = x + 2\,S(x, t_{\max} - \tau) + \sqrt{2}\,\eta(\tau).$$

Probability-flow ODE, same marginals, deterministic: $\mathrm{d}x/\mathrm{d}\tau = x + S(x, t_{\max} - \tau)$.
Initialisation $x(\tau{=}0) \sim \mathcal{N}(0, I)$; termination at $\tau = t_{\max} - t_{\min}$.

The reverse drift follows from the time-reversal theorem for diffusions [27, 28]: for a forward SDE $\mathrm{d}x = f\,\mathrm{d}t + g\,\mathrm{d}W$ the time-reversed process has drift $f - g^2\nabla \log p_t$ and the same diffusion coefficient. With $f = -x$ and $g = \sqrt{2}$ the backward-in-t drift is $-x - 2S$, and rewriting in τ flips the sign to give the box above. Because unit-variance data keep $\mathrm{Var}(x_t) = 1$, initialising at $\mathcal{N}(0, I)$ incurs a bias $e^{-2t_{\max}}$, which is 2.5×10^{-3} at $t_{\max} = 3$. The implementation is `src/reverse.py`, whose module docstring states the same convention; the sampler uses a geometric time grid, denser at small t, for the stiffness reason given in §8.

B The score–posterior identity

Differentiation under the integral. $P_t(x) = \int \mathrm{d}a\, P_0(a)\, g_{\Delta_t}(x - e^{-t}a)$ with $g_{\Delta_t}(u) = (2\pi\Delta_t)^{-n/2} e^{-\|u\|^2/2\Delta_t}$. Since $\nabla_x g_{\Delta_t}(x - e^{-t}a) = -\frac{x - e^{-t}a}{\Delta_t} g_{\Delta_t}(x - e^{-t}a)$ and, for x in a compact neighbourhood, $\|x - e^{-t}a\|\, g_{\Delta_t}(x - e^{-t}a) \le c_1 e^{-c_2\|a\|^2}$ uniformly, the integrand is dominated by an integrable function whenever P_0 has a finite first moment. Dominated convergence then licenses

$$\nabla_x P_t(x) = \int \mathrm{d}a\ P_0(a)\, \frac{e^{-t}a - x}{\Delta_t}\, g_{\Delta_t}(x - e^{-t}a).$$

Dividing by $P_t(x)$ and recognising $P_0(a)g_{\Delta_t}(x - e^{-t}a)/P_t(x) = P(a \mid x, t)$ gives (3). Nothing is approximated. In the empirical-Bayes literature this is Tweedie's formula [7–9].

Risk decomposition. For any u that is a function of x alone,

$$\mathbb{E}\big[\|a - u\|^2 \,\big|\, x\big] = \mathbb{E}\big[\|a - \langle a\rangle_{x,t}\|^2 \,\big|\, x\big] + \|u - \langle a\rangle_{x,t}\|^2, \tag{12}$$

by expanding $a - u = (a - \langle a\rangle_{x,t}) + (\langle a\rangle_{x,t} - u)$ and noting the cross term vanishes because $\langle a\rangle_{x,t} - u$ is x-measurable and $\mathbb{E}[a - \langle a\rangle_{x,t} \mid x] = 0$. Hence $\langle a\rangle_{x,t}$ minimises the conditional risk, with an irreducible floor given by the first term [29]. Optimality is with respect to squared error specifically; a different loss selects a different functional of the posterior. This is why every table in §7 reports error *against* $\langle a\rangle_{x,t}$ rather than raw risk: the floor is common to all estimators and subtracting it is what makes the comparison informative.

C Sum-product on the posterior chain

Posterior factorisation. Bayes' rule applied to (2), with the likelihood factorising sitewise because the forward noise is independent across coordinates, gives (4). Each likelihood factor has degree one among latent variables, so attaching it to the prior's path adds a leaf and cannot create a cycle: the posterior factor graph is a tree. Conditioning usually creates dependence; it does not here, precisely because each observation is generated from a single latent variable. The construction fails the moment the forward noise is correlated across coordinates, since then $P(x \mid a)$ does not factorise, the likelihood becomes a factor touching several a_i, and the posterior graph acquires edges.

Exactness on a tree. Let the graph be a tree and let a_i be a variable node. Removing a_i disconnects it into components, and every factor involves a_i together with variables of exactly one component — a factor touching two components would create a second path between them, hence a cycle. Grouping factors by component gives $P \propto \prod_c g_c(a_i, v^{(c)})$, so conditionally on a_i the components are independent. Unrolling the forward recursion (5),

$$\overrightarrow{m}_i(a_i) \propto \int \mathrm{d}a_{1:i-1}\; \mu(a_1) \prod_{j=2}^{i} K(a_j \mid a_{j-1}) \prod_{j=1}^{i-1} \ell_t(x_j \mid a_j),$$

which is the joint density of everything at or left of site i with the left variables integrated out and $\ell_t(x_i \mid a_i)$ *not* included; symmetrically for $\overleftarrow{m}_i$. The two are conditionally independent given a_i, so their product times the local evidence is proportional to the marginal. The pairwise statement is the same argument with the separator taken to be the edge.

Where the local factor goes. In (5) the local evidence at site $i - 1$ is absorbed *before* the transition, so $\overrightarrow{m}_i$ carries the factors $\ell_t(x_1 \mid a_1), \ldots, \ell_t(x_{i-1} \mid a_{i-1})$ and $\overleftarrow{m}_i$ carries $\ell_t(x_{i+1} \mid a_{i+1}), \ldots, \ell_t(x_n \mid a_n)$, and each appears exactly once in the marginal. Absorbing it *after* would place $\ell_t(x_i \mid a_i)$ in both messages and square it: the belief would be sharper than the truth and every posterior variance too small. This produces plausible-looking output, which is why the convention is pinned by `tests/test_message_normalization.py` rather than left to a comment.

Evidence. Messages are renormalised at each step to prevent underflow; accumulating the discarded constants recovers $\log P_t(x)$ for the discretised model exactly. This is the objective that §5 maximises, and having it in closed form is what makes it possible to check a gradient against a finite difference of the thing being optimised. `src/bp_grid.py` $\rightarrow$ `grid_bp` returns it alongside the beliefs; `src/exact_scores.py` $\rightarrow$ `gaussian_log_evidence` is the closed form it is tested against.

Complexity at the functional level. Each message update integrates one variable against the kernel. Abstractly this is an integral operator applied to a function; represented on N_g points it is one $N_\mathrm{g} \times N_\mathrm{g}$ matrix–vector product, $O(N_\mathrm{g}^2)$. There are $2n$ updates, so one sequence costs $O(nN_\mathrm{g}^2)$ against $O(N_\mathrm{g}^n)$ for naive marginalisation. At $n = 32, N_\mathrm{g} = 401$ that is about 5×10^6 operations. Batching over sequences turns each update into a single GEMM, which is why the same code runs on a GPU without restructuring (`src/backend.py`, gated by `tests/test_backend_parity.py`).

D The Gaussian model and the second-order baseline

For $\varphi = \mathcal{N}(0, q)$ with $q = 1 - \rho^2$ and $a_1 \sim \mathcal{N}(0, 1)$, the vector a is jointly Gaussian with $(\Sigma_0)_{ij} = \rho^{|i-j|}$. Expanding $P_0(a) \propto \exp[-\frac{1}{2q}\sum_{i\geq 2}(a_i - \rho a_{i-1})^2 - \frac{1}{2}a_1^2]$ produces only a_i^2 and $a_i a_{i-1}$ terms, so Σ_0^{-1} is tridiagonal with $(\Sigma_0^{-1})_{ii} = (1 + \rho^2)/q$ in the interior, $1/q$ at the two ends, and $-\rho/q$ on the off-diagonals. Adding the likelihood contributes $(e^{-2t}/\Delta_t)I$, so the posterior precision is tridiagonal at every t: Markovianity is a statement about the precision, while the covariance Σ_0 is dense.

Since $x = e^{-t}a + \sqrt{\Delta_t}z$ with z independent, (a, x) is jointly Gaussian with $\mathrm{Cov}(a, x) = e^{-t}\Sigma_0$ and $\mathrm{Cov}(x) = e^{-2t}\Sigma_0 + \Delta_t I =: \Sigma_t$, whence $\mathbb{E}[a \mid x] = e^{-t}\Sigma_0\Sigma_t^{-1}x$ and, by (3), $S(x, t) = -\Sigma_t^{-1}x$. For zero-mean jointly Gaussian variables the conditional expectation is linear, so it coincides with the LMMSE estimator [30], which proves the second half of Proposition 1. The first half is the observation that both maps in (7) read only the mean and variance of φ, so the whole recursion is a function of $(0, q)$ alone and therefore returns what it returns on the Gaussian chain.

Consequently, on this model, the *message-representation error* and the *model error* coincide: projecting messages onto Gaussians and replacing the prior by its covariance-matched Gaussian give the same estimator. This is a property of single-Gaussian closure with linear transitions, not of Gaussian approximations in general, and the two separate for richer message families.

E Numerical representation

Messages are represented on a uniform grid $u_1, \ldots, u_{N_\mathrm{g}}$ spanning $[-A, A]$ with spacing $h = 2A/(N_\mathrm{g} - 1)$ and composite trapezoidal weights $w_k = h$ for interior points, $h/2$ at the two ends. All recursions run in the log domain with a per-step maximum subtracted before exponentiating, and the removed constants are restored in the evidence, so no message underflows.

Two error sources. *Truncation*: probability mass outside $[-A, A]$ is discarded rather than transported, an error controlled by A and essentially independent of N_g. *Quadrature*: the trapezoidal rule is second order in h for integrands with bounded second derivative, an error controlled by h at fixed A. Figure 2 measures them separately, using the worst-case boundary mass over sites for the first and the interior column-mass residual of the transition matrix for the second; the interior restriction $|u| \leq A/2$ removes columns whose transition density has a tail outside the grid, which is what makes the second diagnostic measure quadrature alone. The observed rates are: boundary mass $3 \times 10^{-4} \to 1.4 \times 10^{-8} \to 1.4 \times 10^{-10}$ at $A = 4, 8, 10$; interior residual falling by a factor of four per halving of h, i.e. $O(h^2)$.

At $A = 4$ the interior residual saturates at $\sim 10^{-3}$ instead of continuing to fall, because at that domain even the "interior" columns have truncated tails — the floor is the truncation error leaking into the quadrature diagnostic, and it is visible in the data (`outputs/exp_18/boundary.csv`). This is the reason the two diagnostics have to be read together and neither alone certifies convergence.

What is not claimed. A grid-resolution sweep at fixed A does not rule out truncation error, and a domain sweep at fixed N_g does not rule out quadrature error. We report both axes and the two diagnostics; we do not claim continuum convergence, only that at $N_\mathrm{g} = 401$, $A = 8$ both diagnostics sit far below every effect the paper reports. For the Laplace family, whose innovation density has a discontinuous derivative at the origin, we claim second order and no better.

F Learning procedure

Objective. Maximum marginal likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(x^\mu)$, with the complete-data objective $\log P_\theta(a, x) = \log \mu(a_1) + \sum_{i\geq 2}\log K_\theta(a_i \mid a_{i-1}) + \sum_i \log \ell_t(x_i \mid a_i)$. Only the middle term depends on θ, which is why Ξ of (10) is sufficient and why $\mathcal{Q}(\theta \mid \theta') = \langle \Xi(\theta'), \log K_\theta\rangle$ up to θ-free constants.

E-step. One forward and one backward sweep per sequence, then a rank-one accumulation of the pairwise statistics: writing $\ell^{(i)}$ for the vector $\ell_t(x_i \mid \cdot)$ on the grid, and with $f_i = w \odot \overrightarrow{m}_i \odot \ell^{(i)}$

and $g_i = w \odot \overleftarrow{m}_i \odot \ell^{(i)}$, the pairwise belief on edge $(i, i+1)$ is $f_i(j)\, K[k,j]\, g_{i+1}(k)/Z_i$ with $Z_i = g_{i+1}^\top K f_i$, and $\Xi = \left(\sum_{\mu,i} g_{i+1}(f_i)^\top / Z_i\right) \odot K$. Ξ sums to the number of edges in the dataset, which is asserted in the tests. Exact for the discretised model, in the sense of Remark 2(ii). Implementation `src/em.py` $\to$ `e_step`.

M-step, by kernel.

- *Gaussian*, $\theta = (\rho, q)$: closed-form maximiser, the belief-weighted Yule–Walker equations $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$. Exact maximisation, hence exact EM for that family.
- *Laplace*, $\theta = (\rho, b)$: closed form despite non-differentiability. Profiling b out of $\mathcal{Q}$ turns the problem into minimising the belief-weighted mean absolute residual, whose exact solution is a weighted median. Exact maximisation.
- *Mixture* (9): the complete-data problem carries a second latent variable, so the M-step is itself an inner EM over the mixture label with Ξ held fixed. We run **four** inner conditional-maximisation sweeps, alternating the (π, ν, s^2) block and the ρ block, both closed form. This increases $\mathcal{Q}$ without maximising it: **generalised EM / ECM** [23], not exact EM.
- *Mixture-density network*: one backtracked Adam step, accepted only if $\mathcal{Q}$ increased, halving the step up to five times otherwise and standing still if no ascent is found. Also generalised EM.

Guarantees, initialisation, stopping. Generalised EM guarantees monotone ascent of $\mathcal{L}$ [22, 24]; convergence is to a stationary point, not necessarily a global maximum. Mixture components are initialised spread over $[-\sqrt{\mathrm{Var}}, \sqrt{\mathrm{Var}}]$ with random scales; ρ is initialised at 0.3 in every experiment against a truth of 0.85. Iteration stops when the relative increase of $\mathcal{L}$ falls below 10^{-9}, capped at 50–120 iterations depending on the experiment. Variances are floored at 10^{-6} to prevent component collapse; no other regularisation is applied, and the component means are unconstrained (§5). Monotonicity is recorded per run as the largest decrease across iterations and is zero in every run reported here.

G Experimental protocol

chain	Laplace AR(1), $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0,1)$, $n = 32$
other families	Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$) all variance-matched to q, so $\mathrm{Cov}(a_i, a_j) = \rho^{\lvert i-j \rvert}$ exactly
grid	$N_{\mathrm{g}} = 401$, $A = 8$ ($h = 0.04$); refinements $N_{\mathrm{g}} \in \{801, 1601\}$
noise levels	$t \in \{0.1, 0.2, 0.4, 0.8, 1.6\}$ for training and pointwise evaluation
budgets	$M \in \{32, 128, 512, 1024, 2048, 4096\}$
test set	256 held-out sequences, seed `exp07-test`, disjoint from all training seeds
replicates	six; the replicate index is mixed into every training seed, so replicates vary the training set *and* the initialisation; the test set is common to all six
EM–BP	mixture-innovation kernel (9), $C = 4$ unless swept; 12 free parameters ρ initialised at 0.3; 50–120 EM iterations; tolerance 10^{-9}
global MLP	fully connected, 25,248 parameters, denoising score matching, 20,000 steps both parameterisations (ε and a) trained; Adam
local CNN	weight-shared one-dimensional convolution, radius r, receptive field $2r+1$
capacity control	24 configurations, 3.2k–297k parameters
hardware	CPU: single-threaded, `medium_cpu` partition, Bocconi HPC GPU: CUDA 12.4, CuPy 13.6, gated by a CPU/GPU parity test before any sweep

Selection rules, stated plainly. There is **no validation split**. Two selections therefore happen on the evaluation set and both favour the baselines: (i) Table 1 reports, at each noise level, the better of the two denoising-score-matching parameterisations, which is oracle post-selection over two models; (ii) the convolution's receptive field is a fixed configuration value read off the same sweep that is evaluated on the test set. The numbers in §7 that depend on (ii) are labelled exploratory. Replacing both with validation-based selection is the first item of the revision plan.

H Reverse-generation protocol

Sampler. Euler–Maruyama on the reverse SDE of Appendix A, from $t_{\max} = 3$ down to $t_{\min} = 0.02$, on a geometric time grid (denser at small t, where the drift stiffens like $1/\Delta_t$). Common random numbers across estimators: the same generator state drives every arm, so paired differences isolate the score and not the noise.

Why the reference is forward-noised. Integration stops at $t_{\min} > 0$, so each arm produces a sample of $P_{t_{\min}}$, not of P_0. Comparing that against clean data measures the stopping point. Concretely, adding independent Gaussian noise of variance v to an innovation of variance q and excess kurtosis κ gives excess kurtosis $\kappa\,(q/(q+v))^2$, since fourth cumulants add and the variance grows by v. At $\rho = 0.85$, $t_{\min} = 0.02$ the residual innovation noise is $v = \Delta_t(1+\rho^2)$ and the predicted target is 1.9098 against a clean 3.0. We therefore draw a reference from P_0 and noise it forward to the same $t_{\min}$: both sides are then samples of the same law and the target follows in closed form. Two earlier versions of this comparison were wrong in exactly the two available ways — comparing against clean data, which measured the stopping floor, and applying a denoising readout, which inverted the bias through shrinkage — and the closed-form target is what identified both.

Metrics. Reported per arm: innovation excess kurtosis with a bootstrap standard error that resamples chains rather than entries; $D_{\mathrm{KL}}(\text{true} \,\|\, \text{generated})$ by histogram; relative Frobenius error of the covariance and the worst error over lags; marginal variance; and, on the pointwise side, MSE against the Bayes denoiser with the Bayes risk floor and the zero-predictor ceiling recorded alongside. Implementation `src/sample_metrics.py`, tested by `tests/test_sample_metrics.py`.

Convergence. Step counts $\{100, 200, 400, 800\}$; see Figure 8. Counts 1600 and 3200 were launched but produced no output and are not reported.

I Additional results

Table 3: Generated innovation excess kurtosis by innovation family, $M = 2048$, $C = 4$, median over three seeds. Families are variance-matched, so they share $\rho^{|i-j|}$ exactly and differ only beyond second moments. The reference column is grid BP under the true kernel, put through the same sampler, and is itself a Monte Carlo estimate. The final column is $|\text{EM–BP} - \text{ref}| - |\text{CNN} - \text{ref}|$: positive means the network is closer. Source `outputs/exp_16/family_*/generation.csv`. Empirical.

family	reference	EM–BP	local CNN	global MLP	disagreement
Gaussian	-0.022	0.019	0.047	0.053	$\mathbf{-0.028}$
mixture	-0.467	-0.040	-0.133	0.030	0.094
uniform	-0.766	-0.056	-0.302	0.018	0.245
Laplace	1.921	0.812	1.273	0.170	0.462
Student	2.940	0.989	1.478	0.191	0.489

Experiments that did not settle their question. Two are worth recording because they constrain what may be claimed. The receptive-field sweep (`exp_12`) measures how error falls with the window radius of a local predictor, which is a proxy for the locality of the score but not a statement about it; without a validation split its selected radius is not defensible, so its numbers are exploratory. The five-family check that the information-form recursion reproduces the closed-form Gaussian posterior mean is asserted in `tests/test_gaussian_bp_equivalence.py` at 10^{-12} for the Laplace chain but is not persisted as an output for the other families, so no aggregate figure is quoted.

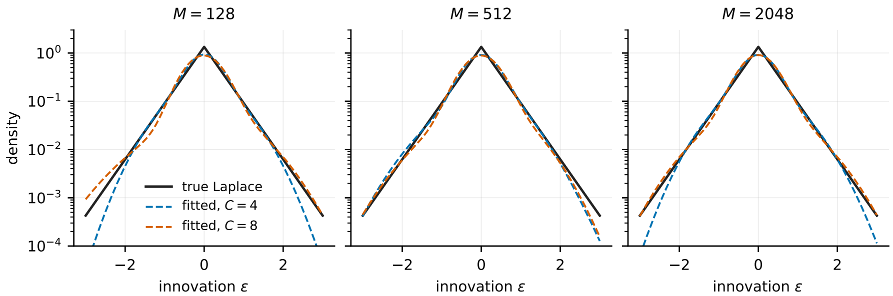

Figure 7: The fitted innovation density against the true Laplace density, at three data budgets and two capacities, on a log scale so the tails are visible. The deficit at $C = 4$ is in the tail, not the bulk, which is the mechanism behind the generative disagreement of §8. Fitted excess kurtosis is 1.63–3.45 at $C = 4$ and 2.83–3.06 at $C = 8$ against a true 3.0; the fitted innovation mean, which is monitored rather than constrained, falls from 1.9×10^{-2} at $M = 128$ to 3.6×10^{-4} at $M = 2048$. Source `outputs/exp_18/innovation_density.csv` and `innovation_summary.csv`. Empirical.

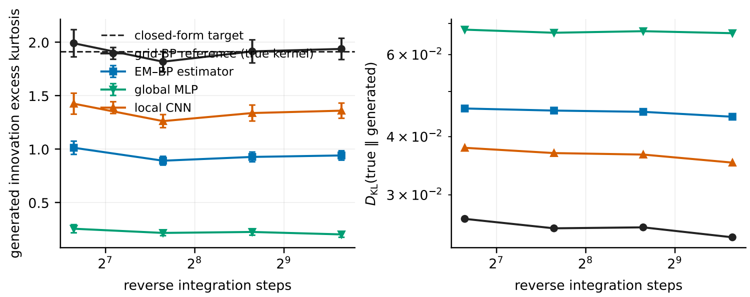

Figure 8: Reverse-sampler convergence. Generated innovation excess kurtosis and $D_{\mathrm{KL}}(\text{true} \,\|\, \text{generated})$ against the number of Euler–Maruyama steps, all three arms; error bars are bootstrap standard errors resampling chains. The reference arm sits within one standard error of the closed-form target at every step count and the arm ordering does not change, so the comparison of §8 is not an artefact of one discretisation over this range. Source `outputs/exp_16/calibrate_steps*/steps.csv`. Empirical.

J Reproducibility map

claim	script (`experiments/`)	output (`outputs/`)
grid and domain convergence	`exp_01_grid_validation`	`exp_01_.../grid_heatmap.csv`
boundary and quadrature diagnostics	`exp_18_... --parts boundary`	`exp_18/boundary.csv`
closure error vs t	`exp_02_laplace_gaussian_...`	`exp_02_.../laplace_summary.csv`
closure error by family	`exp_03_nongaussian_...`	`exp_03_.../innovation_sweep.csv`
EM monotonicity, ρ recovery	`exp_18_... --parts emtrace`	`exp_18/em_trace.csv`
learned innovation density	`exp_18_... --parts density`	`exp_18/innovation_density.csv`
parameter-recovery rate	`exp_06_em_parameter_...`	`exp_06_.../em_rate.csv`
Fisher information vs t	`exp_06_em_parameter_...`	`exp_06_.../price_of_noising.csv`
Fisher identity vs finite diff.	`exp_08_gradient_vs_...`	`exp_08_.../gradient_vs_exact.csv`
sample efficiency (Tab. 1)	`exp_07_em_vs_score_...`	`replicates/merged_summary.csv`
network capacity control	`exp_07_em_vs_score_...`	`exp_07_.../capacity.csv`
inference and training cost	`exp_07_em_vs_score_...`	`exp_07_.../{inference,training}_cost.csv`
receptive field (exploratory)	`exp_12_receptive_field`	`exp_12_.../efficiency_three_way.csv`
sampler step convergence	`exp_16_... MODE=calibrate`	`exp_16/calibrate_steps*/steps.csv`
generation by budget	`exp_16_... MODE=generate`	`exp_16/gen_n*_seed*/generation.csv`
generation by family	`exp_16_sampling_validation`	`exp_16/family_*/generation.csv`
capacity: generation	`exp_16_sampling_validation`	`exp_16/components_C*/generation.csv`
capacity: pointwise	`exp_16_sampling_validation`	`exp_16/cpoint_C*/pointwise.csv`
discrete-alphabet variant	`exp_10_discrete_alphabet`	`exp_10_.../discrete_*.csv`

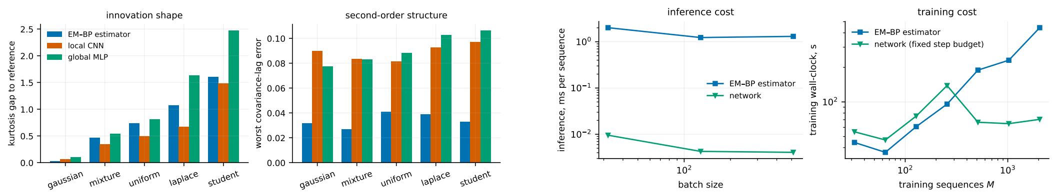

Figure 9: Left: generated-law diagnostics across the five innovation families at $C = 4$ — gap to the reference arm's kurtosis, and worst covariance-lag error. The Gaussian family is the negative control of Table 3. Right: measured cost. Inference is 210–289 times slower per sequence than a forward pass of the network; training cost grows linearly in M for the estimator and is fixed by the step count for the network. Sources `outputs/exp_16/family_*/generation.csv`, `outputs/exp_07_em_vs_score_network/{inference,training}_cost.csv`.

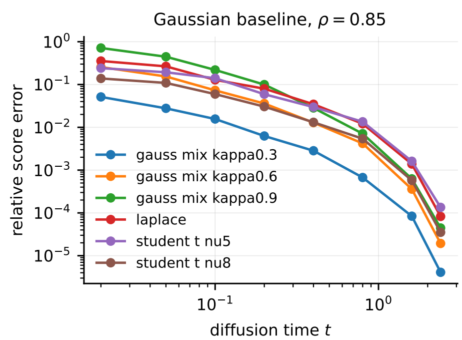

Figure 10: Error of the Gaussian second-order baseline against grid BP under the true kernel, by innovation family, at $\rho = 0.85$. The ordering by innovation excess kurtosis is visible and the decay with t is common to all families. Source `outputs/exp_03_nongaussian_innovation_sweep/innovation_sweep.csv`.

Every figure in this note is produced by `paper/figures/make_figures.py`, which reads the outputs above and fails loudly if one is missing; no number in any figure is entered by hand. Exact commands, environment and HPC job configuration are in `REPRODUCIBILITY.md`, and the claim-level audit behind this revision is `REVISION_AUDIT.md`.